WORKED SOLUTION

STEP II 2012, Question 6

Pure • Euclidean geometry

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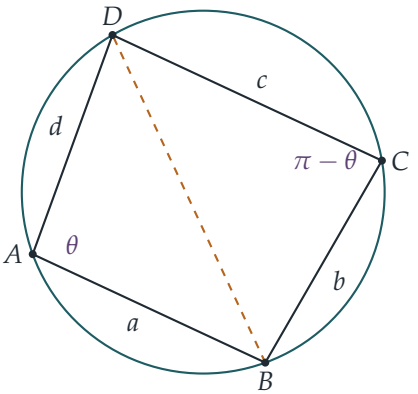
Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s2-q6

The key idea

The diagonal BD splits the quadrilateral into two triangles, and BD can be written in two ways with the cosine rule. What makes a cyclic quadrilateral special is that opposite angles add up to π, so the angle at C is $\pi - \theta$. Then $\cos C = -\cos\theta$ and $\sin C = \sin\theta$, and both the cosine rule and the area formula $\frac{1}{2}xy\sin(\text{angle})$ involve only the single angle θ. After that, $\sin^2\theta + \cos^2\theta = 1$ and some careful factorising do the rest.

Solution



Finding $\cos\theta$ and $\sin\theta$

Since $ABCD$ is cyclic, the opposite angles DAB and BCD sum to π, so $\angle BCD = \pi - \theta$. Hence

$$\cos\angle BCD = -\cos\theta, \qquad \sin\angle BCD = \sin\theta.$$

Apply the cosine rule to the diagonal BD, first in triangle ABD (sides a, d, included angle θ) and then in triangle CBD (sides b, c, included angle $\pi - \theta$):

$$BD^2 = a^2 + d^2 - 2ad\cos\theta = b^2 + c^2 + 2bc\cos\theta.$$

Collecting the $\cos\theta$ terms gives $2(ad + bc)\cos\theta = a^2 + d^2 - b^2 - c^2$, so

$$\cos\theta = \frac{a^2 + d^2 - b^2 - c^2}{2(ad + bc)}.$$

The diagonal BD divides the quadrilateral into these two triangles, so their areas add to Q:

$$Q = \tfrac{1}{2}ad\sin\theta + \tfrac{1}{2}bc\sin(\pi - \theta) = \tfrac{1}{2}(ad + bc)\sin\theta, \qquad \text{so} \qquad \sin\theta = \frac{2Q}{ad + bc}.$$

(We may divide by $ad + bc$ because all the side lengths are positive.)

The formula for $16Q^2$

Substitute both expressions into $\sin^2\theta + \cos^2\theta = 1$:

$$\frac{4Q^2}{(ad + bc)^2} + \frac{(a^2 + d^2 - b^2 - c^2)^2}{4(ad + bc)^2} = 1.$$

Multiplying through by $4(ad + bc)^2$,

$$16Q^2 = 4(ad + bc)^2 - (a^2 + d^2 - b^2 - c^2)^2,$$

as required.

Brahmagupta's formula

The right-hand side is a difference of two squares, $X^2 - Y^2$ with $X = 2(ad + bc)$ and $Y = a^2 + d^2 - b^2 - c^2$.

? **Ask yourself** How do we turn $X - Y$ and $X + Y$ into products of linear factors?

Group the terms so that each is again a difference of two squares. Put the a, d terms together and the b, c terms together:

$$X - Y = (b^2 + 2bc + c^2) - (a^2 - 2ad + d^2) = (b + c)^2 - (a - d)^2,$$
$$X + Y = (a^2 + 2ad + d^2) - (b^2 - 2bc + c^2) = (a + d)^2 - (b - c)^2.$$

Factorising each of these again,

$$\begin{aligned} 16Q^2 &= (X - Y)(X + Y) \\ &= (b + c - a + d)(b + c + a - d)(a + d - b + c)(a + d + b - c). \end{aligned}$$

With $2s = a + b + c + d$, each bracket is $2s$ minus twice one side: for example $b + c + d - a = 2s - 2a$. So the four brackets are $2(s - a)$, $2(s - d)$, $2(s - b)$ and $2(s - c)$, and

$$16Q^2 = 16(s - a)(s - b)(s - c)(s - d), \qquad \text{that is,} \qquad Q^2 = (s - a)(s - b)(s - c)(s - d).$$

The area of a triangle

Every triangle is cyclic, since its three vertices lie on its circumcircle. Now let the side DA shrink to nothing, with D sliding along the circle into A. Then $ABCD$ becomes triangle ABC with sides a, b, c, and $d = 0$. Both sides of the formula depend continuously on D, so it still holds in the limit. With $s = \frac{1}{2}(a+b+c)$, the area Δ of a triangle satisfies

$$\Delta^2 = s(s-a)(s-b)(s-c), \qquad \text{so} \qquad \Delta = \sqrt{s(s-a)(s-b)(s-c)}.$$

This is Heron's formula.

Checks

- A unit square is cyclic with $s = 2$, and $(2-1)^4 = 1$ is its area. For a 3–4–5 triangle, $s = 6$ and $\sqrt{6 \cdot 3 \cdot 2 \cdot 1} = 6 = \frac{1}{2} \cdot 3 \cdot 4$.
- For a rectangle, $a = c$ and $b = d$, so $\cos\theta = 0$, which is right because every angle of a rectangle is $\frac{\pi}{2}$.
- We placed 20 000 random cyclic quadrilaterals on circles and compared the formulas for $\cos\theta$, $\sin\theta$ and Q^2 with the actual angle and the shoelace area. They agreed to within 3×10^{-12}. Heron's formula matched 1000 random triangles.

Common slips

- Forgetting that $\cos(\pi - \theta) = -\cos\theta$. That turns the $+2bc\cos\theta$ into a minus and spoils everything that follows.
- Trying to expand $4(ad+bc)^2 - (a^2+d^2-b^2-c^2)^2$ in full. Factorising it as a difference of squares, twice, is far quicker and much less error-prone.

Beyond STEP

For a general (not necessarily cyclic) quadrilateral, Bretschneider's formula gives the area. It subtracts a term $abcd\cos^2\frac{\alpha+\gamma}{2}$, where α and γ are a pair of opposite angles, so for fixed side lengths the cyclic quadrilateral has the largest possible area. The factorised form of Heron's formula also shows at once that the area vanishes exactly when the triangle inequality becomes an equality. For more geometry in this spirit, see Coxeter and Greitzer, *Geometry Revisited*.

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